

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-014-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
WPS-RNC-PN1**



WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **KWC-06A0017-WPS-RNC-PN1**

Date **23-Feb-13**

Revision No. **1**

Supporting PQR No.(s) **KWC-06A0017-PQR-RNC-PN1**

Date **23-Feb-13**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding + Shield Metal Arc welding**

(Automatic, Manual, Machine, or Semi-Auto.)

(GTAW + SMAW)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal

☐ Nonmetallic ☐ Other

Notes

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1 or 2** to P-No. **1**

Group No. **1 or 2**

Material Specification Type and Grade

A106 Gr.B, A234 WPBEquivalents

TO

A106 Gr.B, A234 WPBEquivalents

Chem. Analysis and Mech. Prop.

Thickness Range :

Base Metal :

Groove

1.6 mm -19.97 mm.

Fillet

All

Pipe Dia. Range :

Groove

NPS 2⁷/8"-Unlimited

Fillet

All

Other

FILLER METALS (QW-404)

Spec. No. (SFA)

GTAW

5.18

SMAW

5.1

AWS No. (Class)

ER 70S-G

E 7016

F-No.

6

4

A-No.

1

1

Size of Filler Metals

Ø 2.4 mm

Ø 3.2 & 4.0 mm.

Deposited Weld Metals

3.0 mm.

7.97 mm.

Thickness Range :

Groove

5.54 mm Max.

14.40 mm. Max.

Fillet

ALL

ALL

Electrode-Flux (Class)

-

-

Flux Trade Name

-

-

Consumable Insert

-

-

Other

Kobelco TGS-50 or Equivalents

Kobelco LB 52 or Equivalents

POSITION (QW-405)

Position(s) of Groove

ALL

Welding Progression

Up

X

Down

Position(s) of Fillet

ALL

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range

N/A

Time Range

N/A

Notes **T>20 mm. heat treatment 593-649 Holding 1 hr**

PREHEAT (QW-406)

Preheat Temp.

Min.

T < 25 mm. = 10°C

T ≥ 25 mm. = 79°C

Interpass Temp.

Max.

250°C

Preheat Maintenance

N/A

Notes

GAS (QW-408)

Percent Composition

Shielding

Gas(es)

(Mixture)

Flow Rate

Trailing

Backing

Argon

99.99%

8-12 Ltr/Min

-

-

-

-

-

-

WPS No. KWC-06A0017-WPS-RNC-PN1 Rev. 1

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☒ AC ☒ DC Polarity EN(GTAW)+ EP(SMAW)
 Amps (Range) 94-150(GTAW) + 100-112(SMAW) Volts (Range) 10-12(GTAW)+ 23-24(SMAW)
 Tungsten Electrode Size and Type Ø 2.4 mm , Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

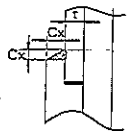
String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE / MULTIPLE ELECTRODE
 Travel Speed (Range) 4.73-12 (GTAW)+5-15.24(SMAW)Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER 70S-G	2.4	DC EN	94-110	10-11	4.73-12	
2nd	GTAW	ER 70S-G	2.4	DC EN	120-150	10-13	10-12	
3 & Over	SMAW	E 7016	3.2 & 4.0	DC EP/AC	100-112	23-34	5-15.24	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of ASME-IX, 2010 latest Revision.

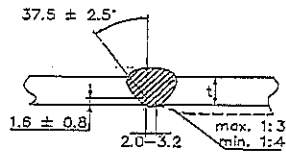
Prepared by: Mr. Chatchawan S. [Signature] 23/02/13
 Name Signed Date
 Reviewed by: ภูชิน สีหอมไชย [Signature] 11/03/13
 Name Signed Date
 Approved by: ภูชิน สีหอมไชย [Signature] 11/03/13
 Name Signed Date

ATTACHED FIGURE JOINT SKETCH



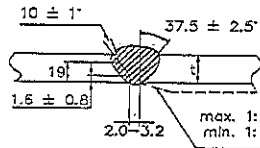
Cx (min.) = $1.25t$ but not less than 3.2 mm.

JOINT No. A1



Misalignment (max.) = 1.5 mm.

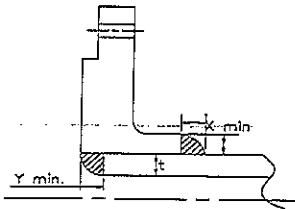
JOINT No. B1 & B2



Approx. 1.6 mm.

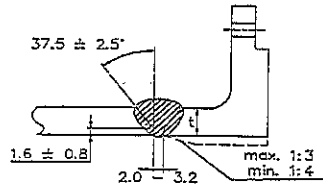
X min. = The lesser of $1.4t$ or thickness of the hub.

JOINT No. A2



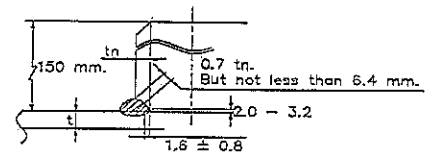
X min. = The lesser of $1.4t$ or thickness of the hub.
Y min. = The lesser of t or 6.4 mm.

JOINT No. A3



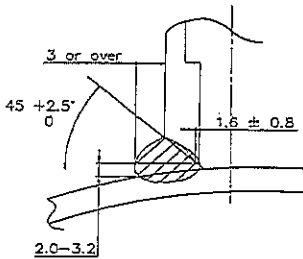
Misalignment (max.) = 1.5 mm.

JOINT No. B3

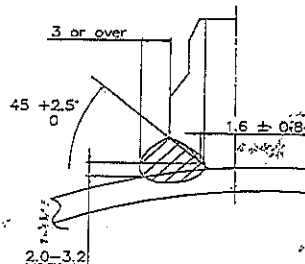


Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

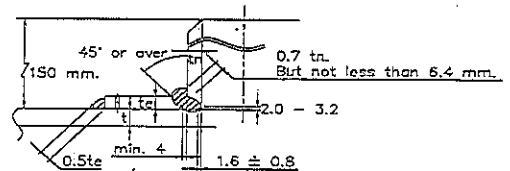
JOINT No. B4



JOINT No. C1

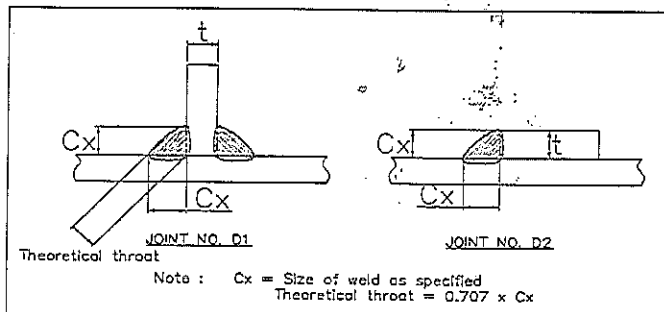


JOINT No. C2



Misalignment (max.) = The lesser of 3.2 mm. or $0.5tn$.

JOINT No. B5



Note: Cx = Size of weld as specified
Theoretical throat = $0.707 \times Cx$

Procedure Qualification Record (PQR)

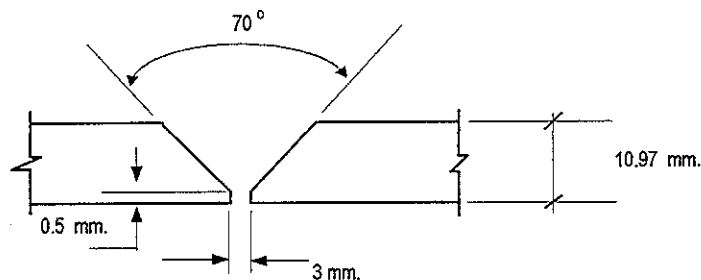
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : KWC-06A0017-PQR-RNC-PN1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	KWC-06A0017-PQR-RNC-PN1	Date :	23 February 2013
WPS Number :	KWC-06A0017-WPS-RNC-PN1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW) + Shield Metal Arc Welding (SMAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)							
Material Spec.		A 106	To	A 106		Temperature		N/A			
Type or Grade		Gr.B		To	Gr.B		Hold Time		N/A		
P. No.		1		To	1		Others				
Thickness of Test Coupon				10.97 mm.				GAS (QW-408)			
Diameter of Test Coupon				Ø 6"							
Others											
FILLER METALS (QW-404)		GTAW		SMAW		Percent Composition					
							Gas (es)	Mixture	Flow Rate		
						Shielding	Argon	99.99%	8 L/Min.		
						Trailing	-	-	-		
						Backing	-	-	-		
						ELECTRICAL CHARACTERISTICS (QW-409)					
SFA Specification		A 5.18		A 5.1		Current		DC (GTAW) DC (SMAW)			
AWS Classification		ER70S-G		E7016		Polarity		EN EP			
Filler Metal F-No.		6		4		Amperage		94 - 150 (GTAW) Voltage 10 - 12 (GTAW)			
Weld Metal Analysis A-No.		1		1				100 - 112 (SMAW) 23 - 24 (SMAW)			
Size of Filler Metal		Ø 2.4 mm.		Ø 3.2 mm.		Tungsten Electrode Size		Ø 2.4 mm. 2% thoriated.EWTh-2			
Others		Kobelco TGS-50		Kobelco LB-52		Others		N/A			
Weld Metal Thickness		-									
POSITION (QW-405)						TECHNIQUE (QW-410)					
Position of Groove			6G			Travel Speed		4.73 - 15.24 cm/min.			
Progression of Welding (Uphill or Down)			Up Hill			String or Weave Bead		Both			
Others			N/A			Oscillation		N/A			
PREHEAT (QW-406)						Multiple or single Pass		Multiple Pass			
Preheat Temperature			37° C			Multiple or Single Electrodes		Multiple Electrodes			
Interpass Temperature (max.)			188° C			Others		N/A			
Others			N/A								

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- KWC-06A0017-PQR-RNC-PN1

Tensile Test Report No.TS-13-02-084

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
545/13 TR1	19.05	9.78	186.31	97.06	520.97	OUT OF WELD
545/13 TR2	19.02	9.88	187.92	97.62	519.46	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-13-02-033

Type and Figure Number	Type of Bend	Indication	Results
545/13 BF1	Transverse face bend	No Open Discontinuity	Accepted
545/13 BF2	Transverse face bend	No Open Discontinuity	Accepted
545/13 BR1	Transverse root bend	2.12,1.57,1.78 (WELD ZONE)	Accepted
545/13 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

Type of Test Radiographic examination results Accepted Report No.WQT-009

Hardness Testing Report No. HV-13-02-015

Macrostructure Report No. MA-13-02-049

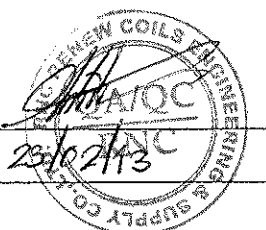
Impact Testing Report No.IM-13-02-092

Welders Name Mr.Prathom M. Welder Number W-009
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



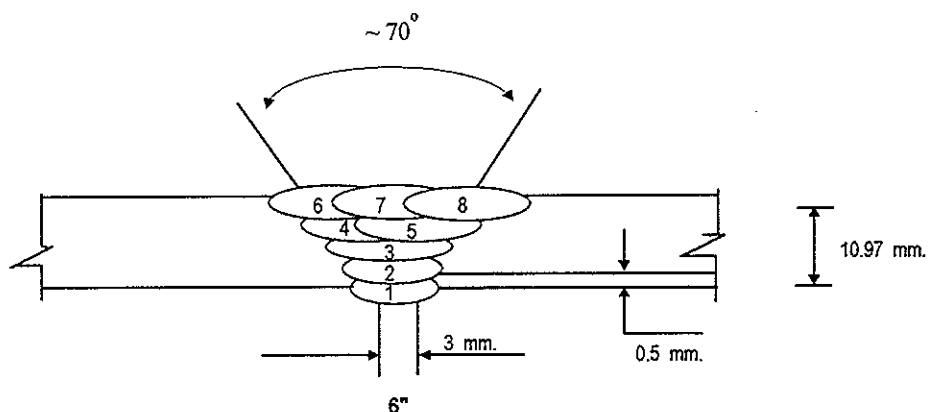
Inspected by : Mr.Wichit K.

Certified by : Mr.Chaloemkiet R.

Date : 22 February 2013

	WELDING PROCEDURE	PQR No. : KWC-06A0017-PQR-RNC-PN1
		WPS No. : KWC-06A0017-WPS-RNC-PN1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST CONDITION AND RESULT

Welding Sequences for KWC-06A0017-PQR-RNC-PN1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	94	10	DCEN	-	-	4.73	11.93	8	-
2	GTAW	ER70S-G	2.4	150	12	DCEN	-	-	8.88	12.17	8	-
3	SMAW	E7016	3.2	100	23	DCEP	-	-	11.71	11.79	-	-
4	SMAW	E7016	3.2	110	24	DCEP	-	-	15.24	10.40	-	-
5	SMAW	E7016	3.2	111	24	DCEP	-	-	9.51	16.81	-	-
6	SMAW	E7016	3.2	110	24	DCEP	-	-	11.68	13.57	-	-
7	SMAW	E7016	3.2	112	24	DCEP	-	-	9.51	16.96	-	-
8	SMAW	E7016	3.2	111	24	DCEP	-	-	7.39	21.63	-	-



PAE TECHNICAL SERVICE CO.,LTD

NDT, TESTING, INSPECTION & ENGINEERING CONSULTANT

69 SOI.OONNUT 64 (SUKSAMARN) SRINAKARIN RD, SUANLUANG,BANGKOK 10250

Tel : 02) 322-0222, Fax : 02)721-2577, RAYONG (038) 607-402-3

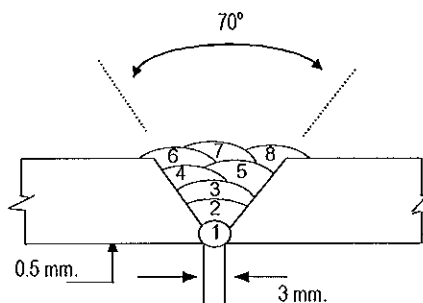
WELDER PROCEDURE QUALIFICATION RECORD

Page No. 1 of 1

Client :	Project Name : PQR	Location : RNC Work Shop	Report No. PQR-017
Renew Coils Engineering & Supply Co.,Ltd.	Customer Order No. : N/A	WPS No. : KWC-0610017-WPS-RNC-PN1	Date : 13-Feb-13

Welder No.	Welder Name	Welding Process	Test Position	Applicable Standard	Material		Electrode Specification	Results	
					Spec	Size		Visual	RT
W-009	MR.PRATHOM M.	GTAW / SMAW	6G	ASME IX	A106 Gr.B	ø 6" Thk. 10.97 mm.	A5.18 / A5.1	ACCEPTED	ACCEPTED

Joint Details



Layer (S)	Process	Filler Metal		Current		Volt	Travel Speed cm/min	Interpas Temp °C	Heat Input (KJ / cm)	Shielding		Backing	
		Class	Dia. (mm.)	Type Polarity	Amp					GAS	Flow Rate (l/min)	GAS	Flow Rate (l/min)
1	GTAW	ER70S-G	2.4	DCEN	94	10	4.73	37	11.93	Argon	8	-	-
2	GTAW	ER70S-G	2.4	DCEN	150	12	8.88	119	12.17	Argon	8	-	-
3	SMAW	E7016	3.2	DCEP	100	23	11.71	123	11.79	-	-	-	-
4	SMAW	E7016	3.2	DCEP	110	24	15.24	143	10.40	-	-	-	-
5	SMAW	E7016	3.2	DCEP	111	24	9.51	150	16.81	-	-	-	-
6	SMAW	E7016	3.2	DCEP	110	24	11.68	153	13.57	-	-	-	-
7	SMAW	E7016	3.2	DCEP	112	24	9.51	180	16.96	-	-	-	-
8	SMAW	E7016	3.2	DCEP	111	24	7.39	188	21.63	-	-	-	-

PAE Inspection By

Witness By

Approved By

SIGNATURE :

NAME :

DATE :

Mr.Wichit K.

14-Feb-2013

SIGNATURE :

NAME :

DATE :

Mr. Chatchawan S.

19/02/13

SIGNATURE :

NAME :

DATE :

ภูชิน สีหอมไชย

Phuchin Seehomchai



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

29780

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. TS-13-02-084

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Reduced Section Tension
Dimension	Ø 6" (10.97 mm Thk.)	Equipment's Capacity	100 TONS
Date Received	19 February 2013	Equipment / Serial No.	INSTRON / 1000DXR1414
Date Tested	20 February 2013	Ambient Temperature	23 ± 5 °C

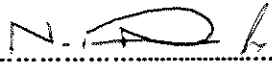
Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	545/13 TR1	N/A	9.78	19.05	186.31	N/A	N/A	N/A	97.06	520.97	N/A	OUT OF WELD
2	545/13 TR2	N/A	9.88	19.02	187.92	N/A	N/A	N/A	97.62	519.46	N/A	OUT OF WELD

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

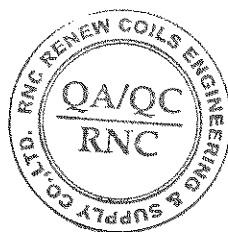
Welder No. : W-009

Reported by : 

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by : 

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

LPN PLATE MILL

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HQ-F-QA-41604

Rev.001/18 n.w. 2556



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



786

TESTING
No. 0129

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. BD-13-02-033

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASME Sec. IX : 2010 edition
Project No.	N/A	Standart Of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 6" (10.97 mm Thk.)	Diameter of Meudrel	40 mm
Date Received	19 February 2013	Bend Angle	180 Degree.
Date Tested	20 February 2013	Equipment/ Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	545/13 BF1	Transverse face bend	10.01x38.10x162.51 mm	No Open Discontinuity	N/A	-
2	545/13 BF2	Transverse face bend	10.00x38.05x161.70 mm	No Open Discontinuity	N/A	-
3	545/13 BR1	Transverse root bend	10.25x38.05x163.03 mm	Open Discontinuity	WELD ZONE	2.12, 1.57, 1.78 mm
4	545/13 BR2	Transverse root bend	10.02x37.97x161.18 mm	No Open Discontinuity	N/A	-

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

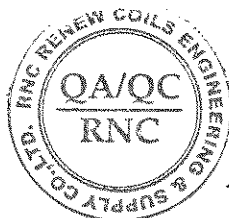
Welder No. : W-009

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnplm.co.th



9792

TESTING
No. 0154

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-13-02-092

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Project Name	WELDER SUPPLY AND PQR	Test Method	ASTM A370-10 and ASTM E23M-07a ⁶¹
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010 edition
Material Specification	A106 Gr.B	Type of Test	Charpy V-Notch Impact
Dimension	Ø 6" (10.97 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Received Date	19 February 2013	Equipment 's Capacity	542 Joules
Tested Date	20 February 2013	Ambient Temperature	23 ± 5 °C

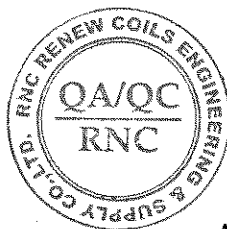
Item No.	Test No.	Test Temperature (°C)	Specimen Dimension			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	545/13-1	-46°C	7.507	10.007	54.81	147.11	199.65	Weld Metal	-
2	545/13-2	-46°C	7.505	10.007	54.79	262.94			
3	545/13-3	-46°C	7.506	10.008	54.86	188.90			
4	545/13-1	-46°C	7.506	10.009	54.92	7.52	9.51	Base Metal	-
5	545/13-2	-46°C	7.506	10.008	54.95	11.32			
6	545/13-3	-46°C	7.506	10.006	54.96	9.70			
7	545/13-1	-46°C	7.507	10.006	54.85	91.68	90.00	HAZ	-
8	545/13-2	-46°C	7.507	10.008	54.88	93.30			
9	545/13-3	-46°C	7.505	10.008	54.78	85.02			

Additional details : Welding Process / Position : GTAW+SMAW / 6G

Welder Name : Mr. Prathom M.

ID No. 3 5301 00149 84 7

Welder No. : W-009



Reported by :

(Mr. Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date 22 / 02 / 13

Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date 22 / 02 / 13

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

2. Partial publicity of the results on testing is prohibited without the written permission from LPN LABORATORY.

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4. This report is tested for LPN Metallurgical Research Center (Thailand) Co., Ltd.

LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

13696

Macrostructure Analysis Report

Page 1 of 1

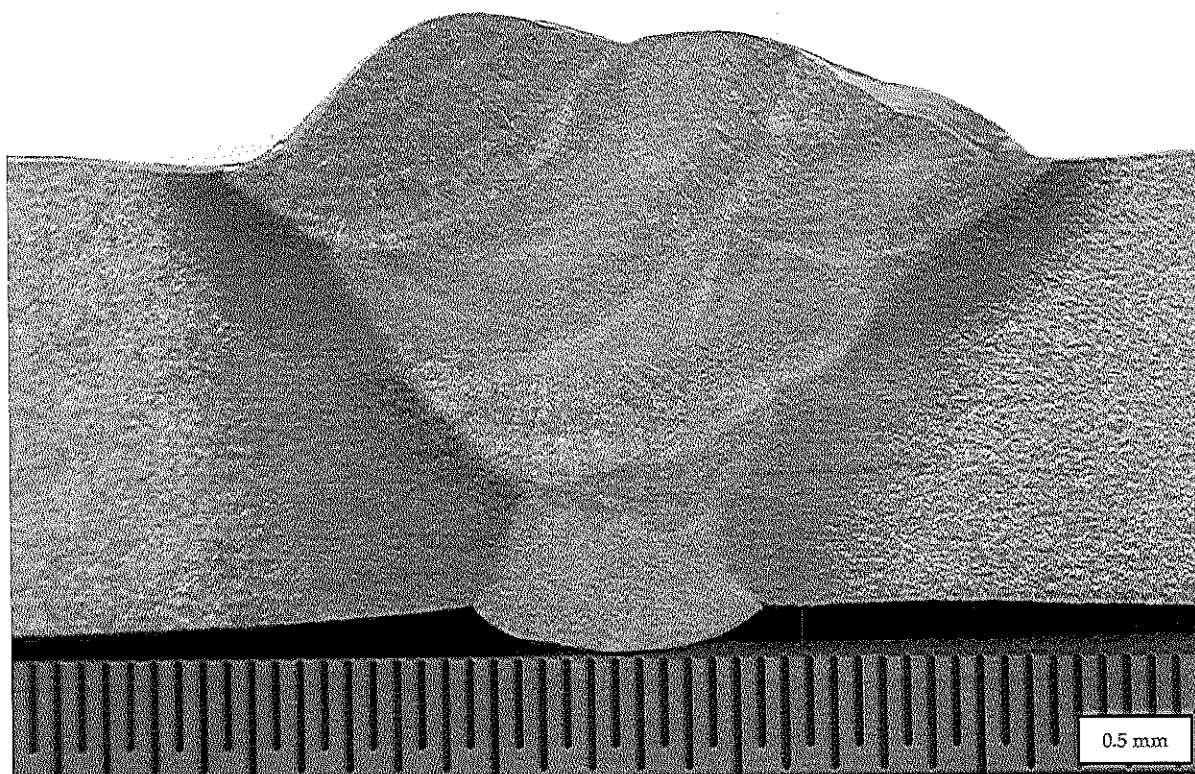
Report No. MA-13-02-049

Revision No.0

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Test No.	545/13 MA	Equipment used	Microscope upto 40X (times)
Test Location	-	Serial No.	LEICA 0650854912
Material Specification	A106 Gr.B	Method of Preparation	ASME Section IX : 2010 edition
Dimension	6" (10.97 mm Thk.)	Test Method	ASME Section IX : 2010 edition
Welding Process/Position	GTAW+SMAW/6G	Standard of Test	ASME Section IX : 2010 edition
Type of Joint	Butt Jiont	Etchant Reagent	25 ml HNO ₃ + 75 ml H ₂ O
Welder Name / No.	Mr. Prathom M. / W-009	Time of Etching	5 minutes
Date Received	18 February 2013	Date Tested	20 February 2013



Macro-etch Examination : The macrostructure shows complete fusion of the weldment. Weld metal and heat affected zone are free of cracks.

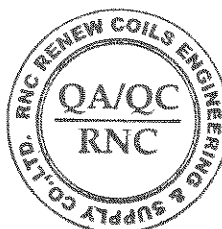
Additional details : -

Reported by :

(Mr.Keerati Sa-ngoen)

(Acting) General Manager - Quality Assurance

Date : 22/02/13



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 22/02/13

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

13232

LPN PLATE MILL

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnpm.co.th

Hardness Testing Report

Customer : RENEWCOILS ENGINEER & SUPPLY CO.,LTD.

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Page 1 of 1

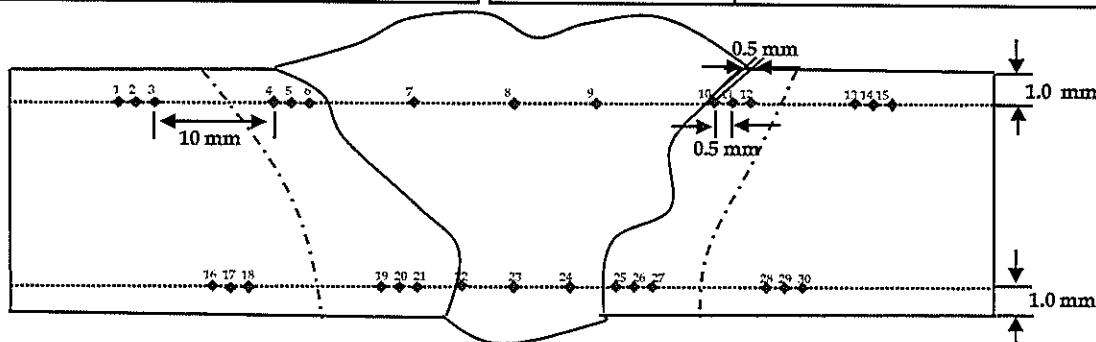
Report No. HV-13-02-015

Revision No. 0

Test Product	WPS No. KWC-06A0017-WPS-RNC-PN1	Material Characterization	Pipe
Test No.	545/13 HN	Test Method	ASTM E92-82 (Reapproved 2003) ^{E2}
Project Name	WELDER SUPPLY AND PQR	Standard of	ASME Section IX : 2010 edition
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A106 Gr.B	Equipment	Wilson Wolpert , 432 SVD (V32D219)
Dimension	6" (10.97 mm Thk.)	Indenter Size	Pyramid diamond, angle of 136°
Welding Process/Position	GTAW+SMAW/6G	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	18 February 2013	Cal. Block	No. A60023 No. A62072
Date Tested	20 February 2013	Ambient Temperature	26.0 °C

Test Location

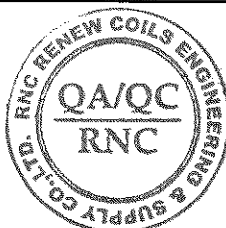
Total = 30 points



Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale HV)	Point No.	Location	Hardness value (Scale)
1	BASE	142.0	16	BASE	148.5	*****		
2	BASE	141.3	17	BASE	147.5			
3	BASE	140.8	18	BASE	148.3			
4	HAZ	147.8	19	HAZ	180.4			
5	HAZ	160.8	20	HAZ	176.7			
6	HAZ	175.0	21	HAZ	176.2			
7	WELD	180.4	22	WELD	170.0			
8	WELD	166.0	23	WELD	164.7			
9	WELD	164.7	24	WELD	167.0			
10	HAZ	180.8	25	HAZ	175.9			
11	HAZ	164.5	26	HAZ	176.9			
12	HAZ	155.9	27	HAZ	178.6			
13	BASE	139.5	28	BASE	143.9			
14	BASE	139.9	29	BASE	141.9			
15	BASE	138.0	30	BASE	142.7			

Additional details :-

Reported by :
(Mr.Keerati Sa-ngoen)
(Acting) General Manager - Quality Assurance
Date : 20/02/13



Approved by :
(Mr.Pavaret Freedawiphat)
Deputy Managing Director
Date : 20/02/13

LPN PLATE MILL

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ArcelorMittal South Africa Limited
Tubular Products
Genl. Hertzog rd.
P O Box 48 Vereeniging 1930
South Africa

MATERIAL TEST CERTIFICATE SEAMLESS TUBE

Telephone +27 (0)16 450 4220
Fax +27 (0)16 423 4906



Licensed under
API Spec 6L and 50T
6L - 0319
50T - 0419



ISO 9001:2000
DE-395997 QM
Test Certificate
EN 10204:2004 TYPE 3.1

ArcelorMittal

Customer: 4000010387
Order No: 040001591051
Certificate Reference No: 040001591051
Product: FULLY KILLED HOT FINISHED CARBON STEEL SEAMLESS TUBES
Specification: ISO3183:2007/API 5L:2007 L245/L290/B/X42 PSL1 ASTM A106.08B/SA53B.07/ A530.04 ASME SA106.08B/SA53B.07/SA530.04
Product Marking: ARCELORMITTAL SA ISO3183/Spec 5L-0319 MONOGRAM 07-10 ASTM/ASME A/SA106B A/SA53B 168.300 (6.626) 10.970 (0.432) 6.000 (19.700) L245/B L290/X42 PSL1
SMLS TESTED 20.7 MPa (3000 psi) CAST NO:89B7130 PROD/O NO: T6641038710

Customer Order/Contract No: SSG7 132443
Material No: 1000000468
Cast/Heat No: 89B7130

Page 1 of 1

General Information

Quantity	Mass	Dimensions			Total Length
		Tube OD	Thickness	Length	
90	22,983.300(kg)	168.300(mm)	10.970(mm)	6.000(m)	540,000(m)
	50,668.983(lb)	6.626(")	0.432(")	19.700 (ft)	1,773.000(ft)

Steel making process	Final Rolling Operation
Electric Arc	Final hot rolling operation finished above 860°C and cooled in still air.

Chemical Composition

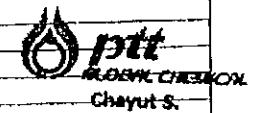
Element(%)	C	Si	Mn	S	P	Cr	Ni	Mo	Cu	V	Al	Ti	Sn	Ca	N	B	Nb	CE
Minimum	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Maximum	0.200	-	1.30	0.030	0.030	0.50	0.50	0.150	0.500	-	-	-	-	-	-	-	-	0.41
Heat	0.170	0.31	0.82	0.008	0.019	0.16	0.08	0.016	0.110	0.005	0.032	0.001	0.008	0.0012	0.0027	0.0003	0.0020	0.36
Product	0.170	0.310	0.820	0.008	0.019	0.160	0.080	0.016	0.110	0.005	0.032	-	-	-	-	-	-	0.356
Product (ADD)	0.170	0.310	0.820	0.008	0.019	0.160	0.080	0.016	0.110	0.005	0.032	-	-	-	-	-	-	0.356

Mechanical Properties

Specification Limits	UTS (Rm)		Yield (0.5%)		% EL 50mm
	MPa	psi	MPa	psi	
Minimum	415	60000	290	42000	30.0
Maximum	-	-	-	-	-
(1) Actual	503	72954	347	50328	38.0
(2) Actual	507	73534	349	50618	38.0
(3) Actual	-	-	-	-	-
(4) Actual	-	-	-	-	-

	UTS (Rm)		Yield (0.5%)		% EL 50mm
	MPa	psi	MPa	psi	
(6) Actual	-	-	-	-	-
(6) Actual	-	-	-	-	-
(7) Actual	-	-	-	-	-
Orientation & type of tensile test	Longitudinal, Strip				
Width of tensile piece (mm)	25.4mm				
Orientation of impact test piece	CERTIFY TRUE COPY				

OTHER TESTS	
Flattening	Passed
Hydrostatic	20700 kPa (3000 psi) - 5 sec
NDI: EMI	PASS - ASTM B570 - 12.5% NOTCH
NDI: UT	UT not required
HV 10 Kg	153 154 156



Remarks:

Material in accordance with NACE MR0175:2003/ISO15156-2:2003, MR0103:2007. Dimensions to ANSI B36.10M-2004
The material conform to the hot yield strength requirements as per ASME, Sect II, Pt D, Table Y-1, 2007 add 2009
All the material conform to the visual and dimensional requirements and is made to a suitable fine grain practice

Quality Assurance Manager: R. Bester

Date of Issue: 2010.08.07

BY URMENA LIMITED
PURCHASE ORDER NO: PO-0045019873
ITEM NO: 6

Certified by: TB

Inspection QA/QC Division
Integrity and Reliability

We hereby certify that the steel grade and quality level of all products are in conformity with the order and comply fully with specification requirements. No changes, amendments or additions may be made to this document. Any changes which are effected shall invalidate this certificate.



20/17 154

KOBELCO

INSPECTION CERTIFICATE

PURCHASER

Date of Issue 11/Aug/2011

Certificate No. TFI1-035

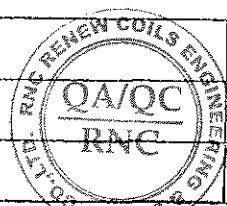
Date of Inspection 16/Aug/2011

TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
LB-52	3.2 x 350	12110	AWS A5.1 E7016

Chemical Composition (%)

Elements	C	Si	Mn	P	S	Ni	Cr	Mo	V	*1*
Weld Metal	0.07	0.52	0.93	0.009	0.003	< 0.01	0.03	< 0.01	0.009	0.97

Tensile Test of Weld Metal			Impact Test of Weld Metal			Tested size for Mechanical test: 4.0x400 mm Tested MFG. No. of Mechanical test: 12009
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Absorbed Energy		
465 MPa	554 MPa	33 %	-30 °C	Avg.	246 J	

Welding Conditions			Postweld Heat Treatment	REMARKS
Type of Current	AC		--- °C x --- Hr.	*1* : Mn+Ni+Cr+Mo+V
Welding Current	120 A			
Welding Voltage	24 V			

1 : Mn+Ni+Cr+Mo+V

We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.



THAI-KOBE WELDING CO., LTD.

CHIEF INSPECTOR

[Signature]



NOBILIT

INSPECTION CERTIFICATE

Date of Issue 17/Feb/2011

Certificate No. KF11-006

Date of Inspection 15/Feb/2011

PURCHASER

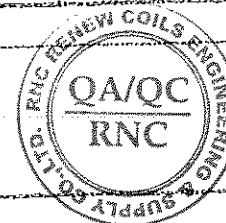
TRADE DESIGNATION	SIZE (mm)	MFG. No.	APPLICABLE SPECIFICATION & CLASSIFICATION
TC-830	2.4	FOB2704	AWS A5.18 ER70S-G

Chemical Composition (%)

Element	C	Si	Mn	P	S	Cu	Ni	Cr	Mo	V
WIRE	0.10	0.73	1.39	0.011	0.017	0.13	0.01	0.01	< 0.01	0.002

Tensile Test of Weld Metal				Impact Test of Weld Metal		Toasted size Yn Mechanics test 2.4 mm Toasted MFG. No. of Mechanics test P9B4140
Yield Strength at 0.2% offset	Tensile Strength	Elongation	Test Temp.	Avg.	Absorbed Energy	
423 MPa	545 MPa	38 %	-30 °C		237 J	

Working Conditions				REMARKS
Electrode	Electrode	Shielding Gas	At	
Electrode	ECEN			
Shielding Gas	Ar			



KOBELCO MIG WIRE (THAILAND) CO., LTD.

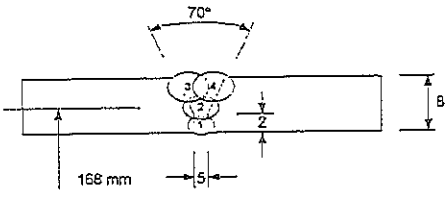


We hereby certify that the test results of the above welding material are as described herein and satisfy the requirements of applicable specification.

CHIEF INSPECTOR *[Signature]*

COPIES

KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Welding Procedure Specification (WPS)

WELDING PROCEDURE SPECIFICATION (WPS)																									
Company Name <u>RENEW COILS ENGINEERING LIMITED PARTNERSHIP</u>	Ref. PQR No. <u>KWC-06A0017-PQR-RNC-PN1</u>																								
WPS No. <u>KWC-06A0017-WPS-RNC-PN1</u>	Date. <u>JULY 20, 2006</u>																								
Revision No. <u>0</u>	Welding Process(es) <u>GTAW and GMAW</u> Type(s) <u>MANUAL</u>																								
(Automatic, Manual, Machine, or Semi-auto)																									
JOINTS (QW - 402) Joint Design <u>6G</u> Backing (Yes) <u> </u> (No) <u>✓</u> Backing Materials (Type) <u>-</u> (Refer to both backing and retainers) ☐ Metal ☐ Nonfusing metal ☐ Non-metallic ☐ Other Sketches, Production Drawings, Weld Symbols or Written Description should show the general arrangement of the parts to be welded. Where applicable, the root spacing and the details of weld groove may be specified. (At the option of the Mfr., sketches may be attached to illustrate joint design, weld layers and bead sequence, e. g., for notch toughness procedures, for multiple process procedures, etc.)	DETAILS 																								
*BASE METALS (QW - 403) P No. <u>1</u> Group No. <u>1</u> To P No. <u>1</u> Group No. <u>1</u> OR Specification type and grade <u>A106 Gr B</u> to Specification type and grade <u>A106 Gr B</u> OR Chem. Analysis and Mech. Prop. <u>-</u> to Chem. Analysis and Mech. Prop. <u>-</u> Thickness Range: Base Metal: Groove <u>3-16 mm</u> / Fillet <u>-</u> Other <u>-</u>																									
*FILLER METALS (QW - 404) Spec. No. (SFA)..... AWS No. (Class)..... F-No..... A-No..... Size of Filler Metals..... Weld Metal Thickness Range: Groove..... Fillet..... Electrode-Flux (Class)..... Flux Trade Name..... Consumable Insert..... Other.....																									
<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td style="width: 50%;">5.18</td> <td style="width: 50%;">5.1</td> </tr> <tr> <td>ER 70S-6</td> <td>E 7016</td> </tr> <tr> <td>6</td> <td>4</td> </tr> <tr> <td>1</td> <td>1</td> </tr> <tr> <td>Ø2.4mm</td> <td>Ø3.2mm</td> </tr> <tr> <td>4 mm (max)</td> <td>12 mm (max)</td> </tr> <tr> <td>4 mm (max)</td> <td>12 mm (max)</td> </tr> <tr> <td>All</td> <td>All</td> </tr> <tr> <td>-</td> <td>-</td> </tr> <tr> <td>-</td> <td>-</td> </tr> <tr> <td>-</td> <td>-</td> </tr> <tr> <td>-</td> <td>-</td> </tr> </table>		5.18	5.1	ER 70S-6	E 7016	6	4	1	1	Ø2.4mm	Ø3.2mm	4 mm (max)	12 mm (max)	4 mm (max)	12 mm (max)	All	All	-	-	-	-	-	-	-	-
5.18	5.1																								
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4 mm (max)	12 mm (max)																								
All	All																								
-	-																								
-	-																								
-	-																								
-	-																								

* Each base metal-filler metal combination should be recorded individually.

(R. Phaonium)
Designed

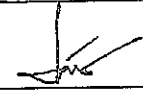

(S. Peansukmanee)
Approved

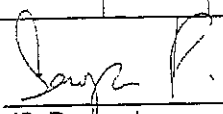
Authorized
Manufacturer

This document is not official unless it carries the authorized signatures.

KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Welding Procedure Specification (WPS) Cont'd

WPS (Back)																																																																																													
WPS No. _____		KWC-06A0017-WPS-RNC-PN1				Rev. <u>0</u>																																																																																							
POSITIONS (QW - 405) Position(s) of Groove <u> All </u> Welding Progression: Up <u> ✓ </u> Down Position(s) of Fillet <u> - </u>				POST WELD HEAT TREATMENT (QW - 407) Temperature Range <u> - </u> Time Range <u> - </u>																																																																																									
PREHEAT (QW - 406) Preheat Temp. Min. <u> - </u> Interpass Temp. Max. <u> 100°C </u> Preheat Maintenance <u> - </u> (Continuous or special heating where applicable should be recorded)				GAS (QW - 408) <table style="width:100%; border-collapse: collapse;"> <thead> <tr> <th></th> <th colspan="3">Percent Composition</th> </tr> <tr> <th></th> <th>Gas(es)</th> <th>Mixture(s)</th> <th>Flow Rate</th> </tr> </thead> <tbody> <tr> <td>Shielding</td> <td><u>Ar</u></td> <td><u>-</u></td> <td><u>20 lpm</u></td> </tr> <tr> <td>Trailing</td> <td><u>-</u></td> <td><u>-</u></td> <td><u>-</u></td> </tr> <tr> <td>Backing</td> <td><u>Ar</u></td> <td><u>-</u></td> <td><u>20 lpm</u></td> </tr> </tbody> </table>						Percent Composition				Gas(es)	Mixture(s)	Flow Rate	Shielding	<u>Ar</u>	<u>-</u>	<u>20 lpm</u>	Trailing	<u>-</u>	<u>-</u>	<u>-</u>	Backing	<u>Ar</u>	<u>-</u>	<u>20 lpm</u>																																																																	
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Backing	<u>Ar</u>	<u>-</u>	<u>20 lpm</u>																																																																																										
ELECTRICAL CHARACTERISTICS (QW - 409) Current AC or DC <u> DC </u> Polarity <u> GTAW DC-, SMAW DC- </u> Amps (Range) <u> GTAW 80, SMAW 100 </u> Volts (Range) <u> GTAW 13.2, SMAW 32.2 </u> (Amps and volts range should be recorded for each electrode size, position, and thickness, etc. This information may be listed in a tabular form similar to that shown below.) Tungsten Electrode Size and Type <u> 2% Thoriated (EWTh-2) </u> (Pure Tungsten, 2% Thoriated, etc.) Mode of Metal Transfer for GMAW <u> - </u> (Spray arc, short circuiting arc, etc.) Electrode Wire feed speed range <u> - </u>																																																																																													
TECHNIQUE (QW - 410) String or Weave Bead <u> GTAW STRING, SMAW WEAVE </u> Orifice or Gas Cup Size <u> Ø10 mm </u> Initial and Interpass Cleaning (Brushing, Grinding, etc.) <u> Brushing and Grinding </u> Method of Back Gouging <u> - </u> Oscillation <u> - </u> Contact Tube to Work Distance <u> - </u> Multiple or Single Pass (per side) <u> MULTIPLE (4) </u> Multiple or Single Electrodes <u> SINGLE </u> Travel Speed (Range) <u> GTAW 4-10 cm/min, SMAW 5-15 cm/min </u> Peening <u> - </u> Other <u> - </u>																																																																																													
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(R. Phaonium)
Designed


(S. Peasukmanee)
Approved


Authorized
Manufacturer

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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Procedure Qualification Records (PQR)

PROCEDURE QUALIFICATION RECORDS (PQR)																								
Company Name <u>RENEW COILS ENGINEERING LIMITED PARTNERSHIP</u>																								
Procedure Qualification No. <u>KWC-06A0017-PQR-RNC-PN1</u>		Date <u>JULY 20, 2006</u>																						
Pre-Qualified WPS No. <u>KWC-06A0017-WPS-RNC-PN1</u>		Revision No. <u>0</u>																						
Welding Process(es) <u>GTAW+SMAW</u>																								
Types (Manual, Automatic, Semi-Auto.) <u>MANUAL</u>																								
JOINTS (QW - 402) Groove Design of Test Coupon (For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used.)																								
BASE METALS (QW - 403) Material Spec. <u>A106</u> Type or Grade <u>Gr. B</u> P No. <u>1</u> to P No. <u>1</u> Thickness of Test Coupon <u>8 mm</u> Diameter of Test Coupon <u>168 mm</u> Other(s) <u>PIPE</u>		POSTWELD HEAT TREATMENT(QW-407) Temperature <u>-</u> Time <u>-</u> Other(s) <u>-</u>																						
FILLER METALS (QW - 404) SFA Specification <u>5.18</u> <u>5.1</u> AWS Classification <u>ER70S-6</u> <u>E7016</u> Filler Metal F-No. <u>6</u> <u>4</u> Weld Metal Analysis A-No. <u>1</u> <u>1</u> Size of Filler Metal <u>Ø2.4mm</u> <u>Ø3.2mm</u> Other <u>-</u> Weld Metal Thickness <u>8 mm</u>		GAS(QW-408) <table border="1" style="width:100%; border-collapse: collapse;"> <thead> <tr> <th></th> <th>Gas(es)</th> <th>Mixture</th> <th>Flow Rate</th> </tr> </thead> <tbody> <tr> <td>Shielding</td> <td><u>Ar</u></td> <td><u>-</u></td> <td><u>20 lpm</u></td> </tr> <tr> <td>Trailing</td> <td><u>-</u></td> <td><u>-</u></td> <td><u>-</u></td> </tr> <tr> <td>Backing</td> <td><u>Ar</u></td> <td><u>-</u></td> <td><u>20 lpm</u></td> </tr> </tbody> </table>			Gas(es)	Mixture	Flow Rate	Shielding	<u>Ar</u>	<u>-</u>	<u>20 lpm</u>	Trailing	<u>-</u>	<u>-</u>	<u>-</u>	Backing	<u>Ar</u>	<u>-</u>	<u>20 lpm</u>					
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POSITION (QW - 405) Position of Groove <u>6G</u> Weld Progression (Up, Downhill) <u>UPHILL</u> Other <u>-</u>		ELECTRICAL CHARACTERISTICS(QW-409) Current <u>DIRECT CURRENT</u> Polarity <u>GTAW DC-</u> <u>SMAW DC-</u> Amp <u>80</u> <u>100</u> Volt <u>13.2</u> <u>32.2</u> Tungsten Electrode Size <u>2.4 mm</u> Other <u>-</u>																						
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KMUTT'S WELDING RESEARCH AND CONSULTING CENTER
Procedure Qualification Records (PQR) Cont'd

PROCEDURE QUALIFICATION RECORDS (BACK)							
TEST RESULTS OF TEST COUPON				PQR No. <u>KWC-06A0017-PQR-RNC-PN1</u>			
TENSILE TESTS (QW - 150)							
Specimen No.	Width	Thickness	Area	Ultimate Total Load	Ultimate Unit Stress	Type of Failure & Location	
	(mm)	(mm)	(mm ²)	(kN)	(MPa)		
PN1-TS01	22.00	4.90	107.80	69.550	645.189	Weld Area	
PN1-TS02	22.90	5.60	128.24	72.350	564.176	Weld Area	
GUIDED BEND TESTS (QW - 160)							
Type & Figure No.		Result			Remarks		
PN1-FB1, Face Bend		No Open Defect at Weldment			-		
PN1-FB2, Face Bend		Open Discontinuity <3.2 mm at Weld			-		
PN1-RB1, Root Bend		No Open Defect at Weldment			-		
PN1-RB2, Root Bend		Open Discontinuity <3.2 mm at Weld			-		
TOUGHNESS TESTS (QW - 170)							
Specimen No.	Notch Location	Specimen Size	Test Temp.	Impact Values			Drop Weight Break (Y/N)
				Ft. Lbs.	Shear	Mils	
Comments: _____							
FILLET WELD TESTS (QW - 180)							
Result-Satisfactory: YES _____ NO _____				Penetration into Parent Metal: YES _____ NO _____			
Macro Result: _____							
OTHER TEST							
Type of Test _____							
Deposit Analysis _____							
Other _____							
Welder's Name _____				Clock No. _____		Stamp No. _____	
Test Conducted by <u>KINGWELD</u>				Laboratory Test No. <u>KWC-06A0017</u>			
We certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirements of Section IX of the ASME Code.							

Eakharat Warinsirurak
(E. Warinsirurak)
Inspector

Authorized
Manufacturer

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